Department of Computer Science

CS CS5707 Artificial Intelligence

Postgraduate Study Guide for 2026/27

TABLE OF CONTENTS

Module Details 1
Access to Support Material and Additional Information 1
Introduction/Aims/Background 1
Learning Outcomes 1
Method of Teaching 2
Lecture/Seminar/Lab Programme 2
Reading List 3
Assessment 4
Late Coursework 4
Deliverables and Feedback – Important Dates 4
Expectations of Artificial intelligence Use 5
Employability and Career Skills 5
Additional Vital Information 5

MODULE DETAILS

Module Leader	Yongmin Li	
Department	Computer Science	
Credits	15	
Other staff	Weibo Liu	
Contact and private study time	Lectures	9 hours
	Labs/Seminars	18 hours
	General Study	123 hours
	Total	150 hours
Assessment	Method	[Proportion of final marks/grade]
	Exam	100%

ACCESS TO SUPPORT MATERIAL AND ADDITIONAL INFORMATION

Most teaching, learning and support material is provided electronically via the University's Brightspace system. Note that the details provided in this study guide are based on the formal module syllabus for this module which sets out the agreed content, learning outcomes, assessment and teaching methods. Module syllabus and scheme of studies documents for your programme of study can be found by on the University's Quality Assurance web pages.

INTRODUCTION/AIMS/BACKGROUND

The aim of the module is to introduce the key concepts, principles and fundamental methods of artificial intelligence, and to develop students' skill in analysing of problem requirements, applying appropriate artificial intelligence methods to defined problems, and evaluating the effectiveness of the adopted approach.

LEARNING OUTCOMES

Whatever module or programme of study you are studying for at Brunel University of London, there are learning outcomes (LO) that you must meet/achieve in order to be awarded the credits which comprise the module and programme of study. In order to get a pass grade (C- or above) in this module, you must meet these learning outcomes below, that is, you must demonstrate ability to:

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LO1: Develop an in-depth knowledge and understanding of the key concepts of artificial intelligence and fundamental methods of artificial intelligence.
LO2: Learn how to apply appropriate artificial intelligence methods to defined problems, justify the approach adopted, and critically evaluate its effectiveness.

METHOD OF TEACHING

This module is intended to be both theoretical and practical in its nature. By engaging with real-world problems, the students will have the best opportunity to learn such techniques and tools. Teaching is organised as a combination of lectures and labs/seminars.

LECTURE/SEMINAR/LAB PROGRAMME

This module has been scheduled to run during Semester 1 and the lectures are scheduled as follows:

Semester 1 [please delete as appropriate]

Week	Lecture Topic	Lecturer
1	Introduction	Yongmin Li
2	Search	Yongmin Li
3	Genetic Algorithms	Yongmin Li
4	Probabilistic Inference	Yongmin Li
5	Fuzzy Systems	Yongmin Li
6	READING WEEK	
7	Image Processing	Yongmin Li
8	Natural Language Processing	Yongmin Li
9	Large Language Models	Yongmin Li
10	Generative AI	Yongmin Li
11		

Labs/Seminars are scheduled as follows:

Semester 1 [please delete as appropriate]

Week	Date	Activity	Labs/Seminars
1		Python and Google Colab	Lab/Seminar
2		Search	Lab/Seminar
3		Genetic Algorithms	Lab/Seminar
4		Probabilistic Inference	Lab/Seminar
5		Fuzzy Systems	Lab/Seminar
6	READING WEEK		
7		Image Processing	Lab/Seminar
8		Natural Language Processing	Lab/Seminar
9		Large Language Models	Lab/Seminar
10		Generative AI	Lab/Seminar
11			

Please note: Week 6 in Semester 1 and week 23 and 26-28 in Semester 2 are weeks during which there are no scheduled lectures, labs, seminars or tutorials. Weeks 15-17 are the Winter Examination Period. Week 18 is the first week of teaching after the Christmas break.

READING LIST

Core reading list

[1] Russell, S. and Norvig, P. (2020). Artificial Intelligence: A Modern Approach. 4th Edition. Pearson Education Limited. (3rd edition 2014 is fine as well)
[2] Negnevitsky M. (2011). Artificial Intelligence: A Guide to Intelligent Systems. 3rd Edition. Addison Wesley.

Supplementary Reading

The core reading list provide mostly the materials for classical AI topics. Additional reading materials will be provided at individual lectures and seminars, especially for modern AI technologies.

Please note that there is a wealth of material, much of it freely available on the web or in the library. The above are suggestions but you are encouraged to search for and make use of other sources.

It is important that you learn to become self-reliant and able to access and assimilate material for yourself. Many of the topics necessary for this project will not be covered by lectures. Therefore, you will need to work through the exercises and guidance material provided in labs, via Blackboard or from your own investigations.

Assessment

The assessment against the learning outcomes specified in this study block is Examination (100%), which assesses your knowledge, understanding and application of all topics covered in the block. The associated assessment block code is CS5807 with the same block name "Artificial Intelligence". The duration of the Examination is 2 hours.

Late Coursework

The clear expectation is that you will submit your coursework by the submission deadline stated in the study guide. In line with the University's Coursework Submission Policy (approved in April 2025), coursework submitted up to 48 hours late will be accepted but the penalties set out in the 'Late Submission' section of the policy will be applied. Work submitted over 48 hours after the stated deadline will automatically be graded NS (non-submission).

Please refer to the Computer Science student information pages and the Coursework Submission Procedure pages for information on submitting late work, penalties applied and procedures in the case of Exceptional Circumstances.

Deliverables and Feedback – Important Dates

You should prepare and submit all coursework according to the instructions in the relevant assessment brief. You should make sure that you are fully aware of the University's policy on plagiarism and collusion. You should also be aware that you *cannot* later claim that you did not know the rules and regulations as you must make yourself familiar with them. If you cannot complete any work on time, you should look at the Department's instructions on what to do. The Department policy is that all coursework must be submitted electronically via the WISEflow system. Please navigate to the Brightspace pages for this module for further details. You will get feedback on your performance via WISEflow for this module. If do not receive your feedback by the given date, you should first contact the module leader. If it proves necessary, you should also contact the Director of Postgraduate Studies.

The important dates:

In the table below, the key dates and tasks associated with the assessment for the module are set out.

Task	Assignment Title	Available on Brightspace	Submission deadline	Feedback due	Weighting (%)
T1	Examination	N/A	N/A	15 working days afterwards	100%

Note: all deadlines are at 11.00 a.m. UK time on the stated date (e.g., if the deadline is 1 February, it means the deadline is 11 a.m. UK time on 1 February)

Expectations of Artificial intelligence Use

The University has general guidance on using artificial intelligence in your studies.

We encourage you to use (or experiment with) AI tools to support your study in this module, especially since this is one of the exciting developments within the subject area of this module. This would undoubtedly help you to develop your understanding of the relevant topics, as well as to know from first-hand experience the advantages and limitations of the AI models behind these tools. However, it is unlikely that copying and pasting responses from these tools will enable you to meet the minimum requirements of the tasks assigned in this module. You will still need to ensure that your solutions actually work to produce the correct results and further analyse these results.

Employability and Career Skills

This module equips you with practical skills in key artificial intelligence techniques, including both classic AI and modern AI algorithms, models, and their applications. Through lectures, seminars and hands-on lab sessions using Python and Google Colab, you will gain critical problem-solving and analytical capabilities essential for high-demand roles of software engineering, data science, and AI research and development.

Additional Vital Information

Information relating to all aspects of your studies, including procedures of how to inform us of problems you are facing with your studies, how to apply for an extension for your coursework submission, plagiarism and types of other academic misconduct, and other important matters, are available on the University's intranet. The Department assumes that you familiarise yourself with this information, so you will need to look at the relevant intranet pages carefully at various times throughout your studies. The Department also operates within the rules and regulations of the University more generally, and you should also look at what are known as 'Senate Regulations' on the University's web pages. These policies and procedures might change from one academic year to another, and it is in your interest to review them on an on-going basis.